



ASTHMA



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INTRODUCTION

Definition

- "a *heterogeneous* disease, usually characterized by chronic airway inflammation. It is defined by the history of respiratory symptoms such as **wheeze, shortness of breath, chest tightness,** and cough that vary over time and in intensity, together with variable expiratory airflow limitation."

  Global Initiative for Asthma (GINA)

- The variable airflow obstruction is often **reversible** either spontaneously or with treatment, although reversibility may **not be complete** in some patients with asthma

  The National Institutes of Health, National Asthma Education and Prevention Program (NAEPP) Expert Panel Report 3 (EPR3)

EPIDEMIOLOGY

- Most common chronic disease among children.
- Adults are **five times** more likely to die from asthma than children, with the highest death rates in **females**
- Approximately 60% of adults and children describe themselves as having **persistent asthma**
- Most patients being diagnosed by 5 years of age
  - 50% of children having symptoms by 2 years of age
- more common in boys

- 30% and 70% of children with asthma will improve markedly or become symptom-free by early **adulthood**
- Chronic disease persists in about 30% to 40% of patients
- More common in adult women than men
- Generally 20% or less develop severe chronic disease
- Predictors of persistent adult asthma
  - atopy, onset during school age, and presence of bronchial hyper-responsiveness (BHR)
- Most **deaths** from asthma occur outside the hospital, and death is rare after hospitalization

ETIOLOGY

- Development of asthma
  - Genetic predisposition plus environmental interaction
- Genetic factors account for 60% to 80% of the susceptibility
  - Not all atopic individuals develop asthma, nor do all patients with asthma exhibit atopy
- Environmental risk factors
  - socioeconomic status, family size,
  - exposure to secondhand tobacco smoke in infancy, and in utero,
  - allergen exposure, ambient air pollution, urbanization, viral respiratory infections including respiratory syncytial virus (RSV) and rhinovirus

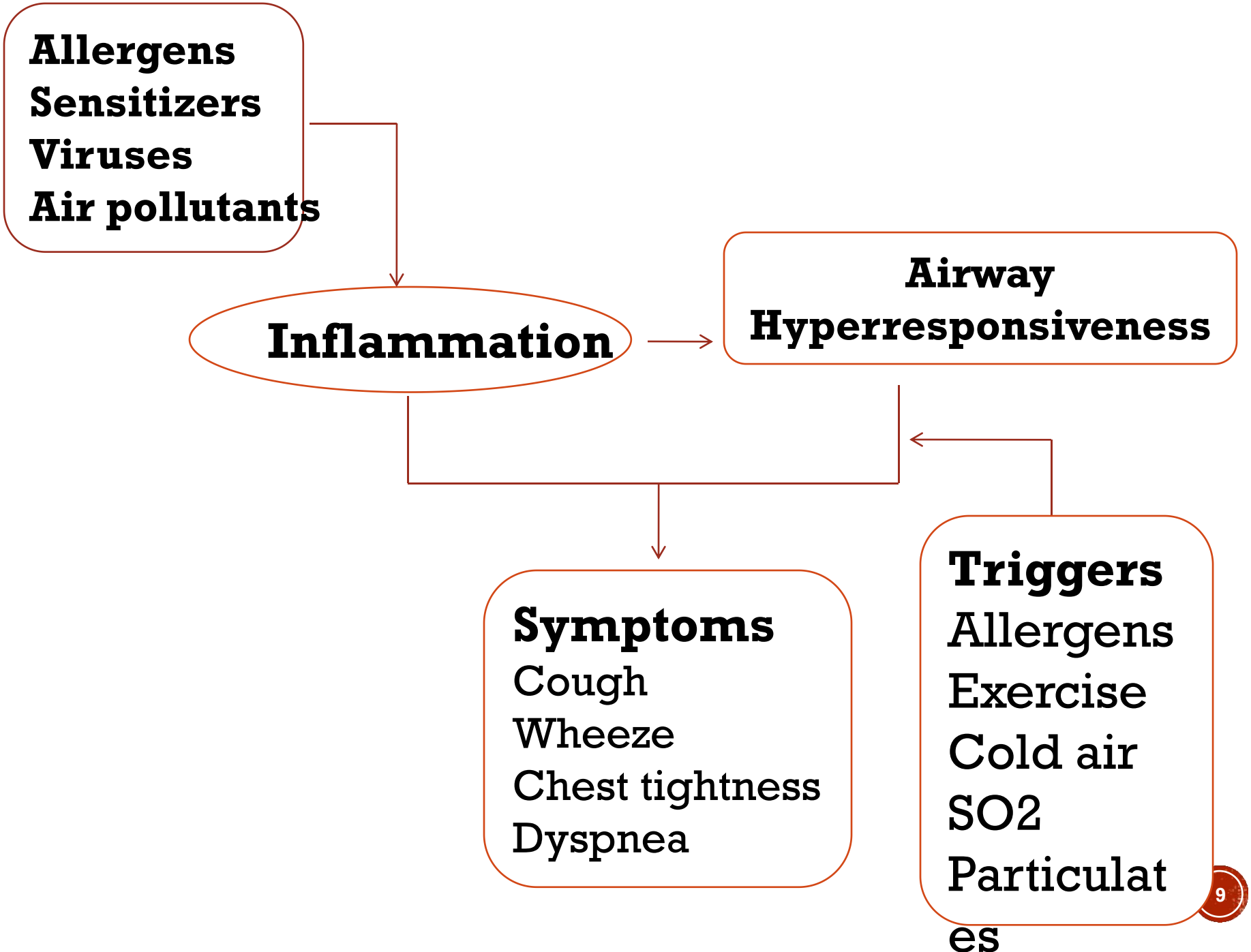
- Environmental **exposures** are the most important precipitants of severe asthma exacerbations.
- Epidemics of severe asthma in cities have followed exposures to high concentrations of **aeroallergens**.
- **Viral respiratory tract infections** remain the single most significant precipitant of severe asthma in **children**
- Other possible factors precipitating exacerbations include air pollution, emotions, exercise, occupational exposures, and drugs.

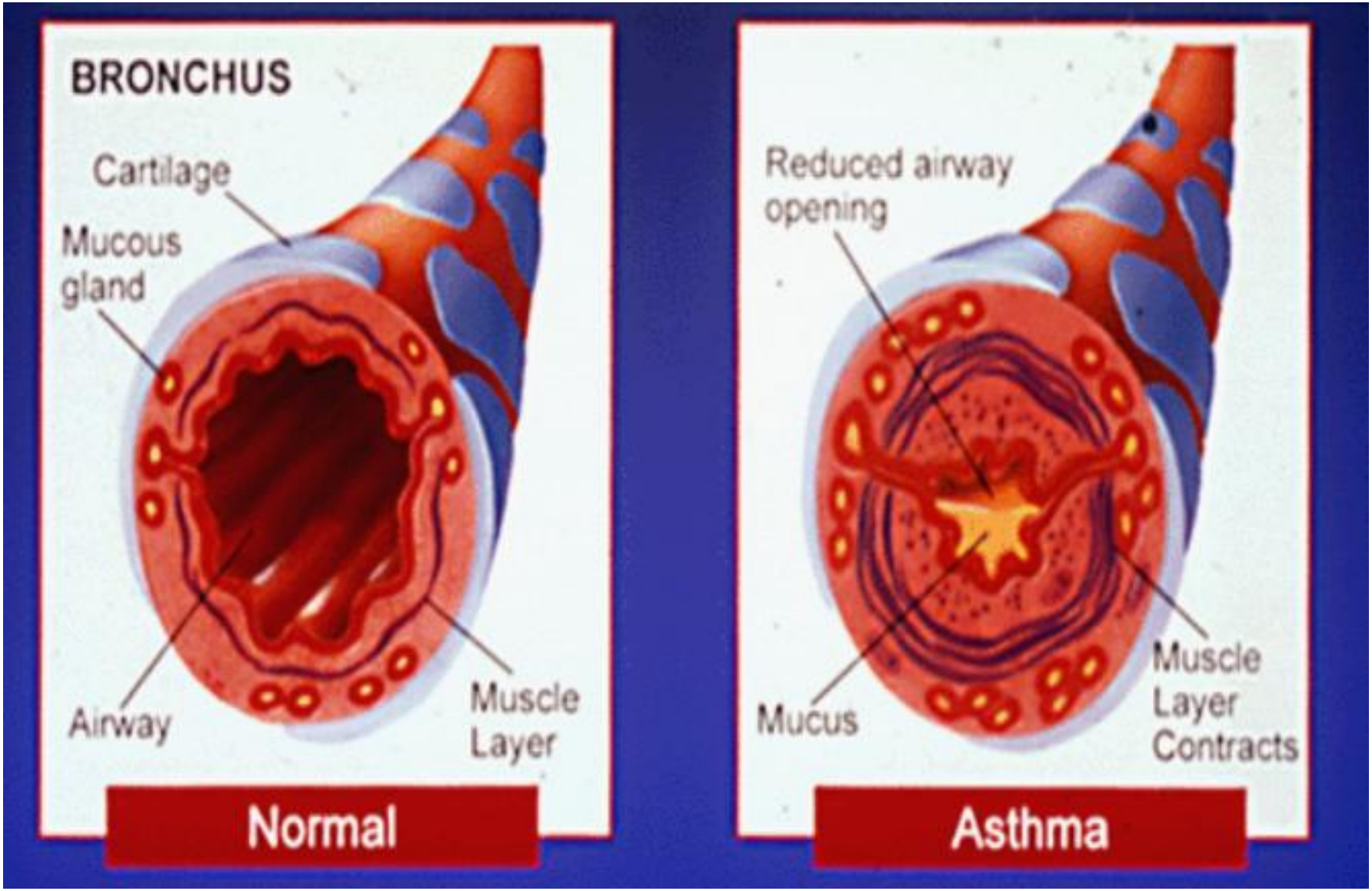
Asthma Triggers

Respiratory infection	rhinovirus, influenza, parainfluenza, *Mycoplasma pneumonia*
Allergens	Airborne pollens (grass, trees, weeds), house-dust mites, animal danders, cockroaches, fungal spores
Environment	Cold air, fog, ozone, sulfur dioxide, nitrogen dioxide, tobacco smoke, wood smoke
Emotions	Anxiety, stress
Exercise	Particularly in cold, dry environments
Drugs / preservatives	Aspirin, NSAIDs (cyclooxygenase inhibitors), sulfites, benzalkonium chloride, nonselective β-blockers
Occupational stimuli	Bakers (flour dust); farmers (hay mold); spice and enzyme workers; printers (arabic gum); chemical workers (azo dyes, anthraquinone, ethylenediamine, toluene diisocyanates, polyvinyl chloride); plastics, rubber, and wood workers (formaldehyde, western cedar, dimethylethanolamine, anhydrides)

PATHOPHYSIOLOGY

- Characteristics of asthma
  - variable degree of airflow obstruction
    - smooth muscle bronchospasm, edema, and mucus hypersecretion
  - Bronchial hyper responsiveness and
  - Airway inflammation


Allergens
Sensitizers
Viruses
Air pollutants
Inflammation
Airway Hyperresponsiveness
Symptoms
Cough
Wheeze
Chest tightness
Dyspnea
Triggers
Allergens
Exercise
Cold air
SO2
Particulates


BRONCHUS
Cartilage
Mucous gland
Airway
Muscle Layer
Normal
Reduced airway opening
Mucus
Muscle Layer Contracts
Asthma

DIAGNOSIS : CLINICAL

- **Episodic asthma**: Paroxysms of wheeze, dyspnea and cough, asymptomatic between attacks.
- **Acute severe asthma**:
  - Upright position, use accessory resp muscles, can't complete sentences in one breath, tachypnea > 25/min, tachycardia > 110/min, PEF < 50% of pred or best, chest hyperresonant, prolonged expiration, breath sounds decreased, inspiratory and expiratory rhonchi.

- **Life-threatening Features**: PEF < 33% of pred or best, silent chest, cyanosis, bradycardia, hypotension, exhaustion, confusion, coma, PaO_2 < 60, PCO_2 normal or increased, acidosis (low pH or high [H+]).
- **Chronic Asthma**: Dyspnea on exertion, wheeze, chest tightness and cough on daily basis, usually at night and early morning; intercurrent acute severe asthma (exacerbations) and productive cough (mucoid sputum), recurrent respiratory infection, expiratory rhonchi throughout and accentuated on forced expiration

DIAGNOSIS : PHYSIOLOGIC

- Demonstration of variable airflow obstruction with reversibility by means of FEV1 and PEF measurement (spirometer and peak flow meter).

1. **FEV1** < 80% of pred – **PEF** < 80% of pred.
2. **Reversibility**: A good bronchodilator response is a 12% or 200ml improvement in FEV1 15 min after inhalation of 200µg salbutamol (2 puffs).
3. **Diurnal peak flow variation**: Normal variation: Morning PEF 15% lower than evening PEF. With asthma this variation is > 15% (morning dipping).

4. *Provocation studies: AHR*

a) **Exercise:** A 15% drop in FEV1 post exercise indicates exercise induced asthma.

b) **Metacholine challenge**: A 20% reduction in FEV1 at Metacholine concentrations < 8mg/ml indicates bronchial hyperreactivity.

CLASSIFICATION OF ASTHMA SEVERITY

			Persistent		
	Components	**Intermittent**	**Mild**	**Moderate**	**Severe**
Impairment	Symptoms	≤2 days/wk	>2 days/wk but not daily	Daily	Throughout the day
	Nighttime awakenings	≤2 x month	3-4 x month	>1 x week, but not nightly	Often 7 x week
	SABA use for Sx control	≤2 days/wk	>2 days/wk, but not >1 x day	Daily	Several times per day
	Interference with normal activity	None	Minor limitation	Some limitation	Extremely limited
	Lung function[a]	FEV_1 >80%	FEV_1 >80%	FEV_1 60%-80%	FEV_1 <60%
		FEV_1/FVC normal	FEV_1/FVC normal	FEV_1/FVC reduced 5% (0.05)	FEV_1/FVC reduced >5% (0.05)
	Exacerbations	**Intermittent**	**Persistent**		
Risk		0-2/yr	>2 in 1 year		
	Recommended step for initiating treatment	Step 1	Step 2	Step 3 and consider short course of oral corticosteroids	Step 4 or 5 and consider course of oral corticosteroid

MANAGEMENT ASTHMA

- Desired Outcomes
- **Chronic Asthma**

(1) Prevent chronic and troublesome symptoms;

(2) Maintain normal or near normal pulmonary function;

(3) Maintain normal activity levels, including exercise and other physical activities;

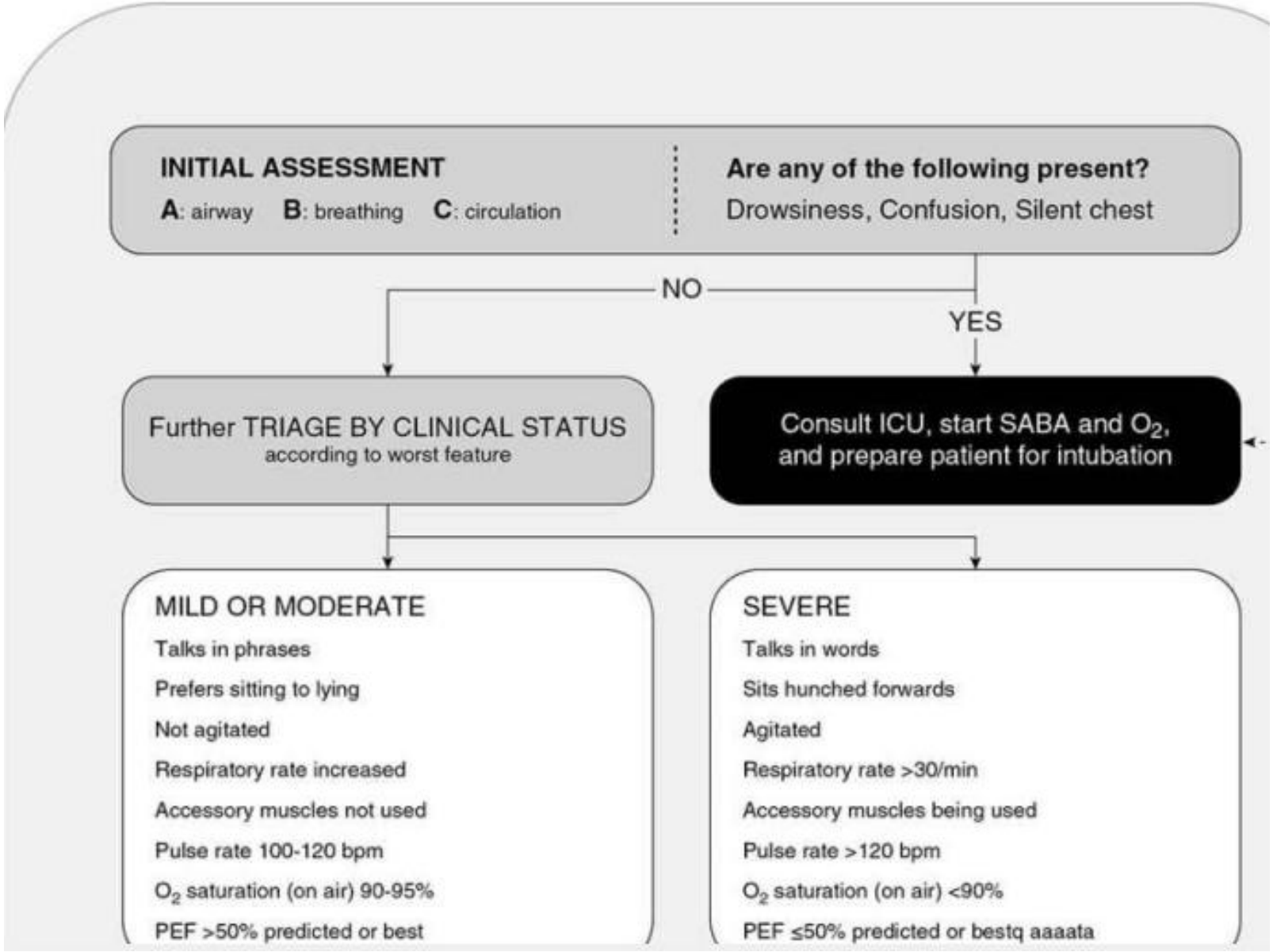
(4) Prevent recurrent exacerbations of asthma and minimize the need for emergency department visits or hospitalizations;

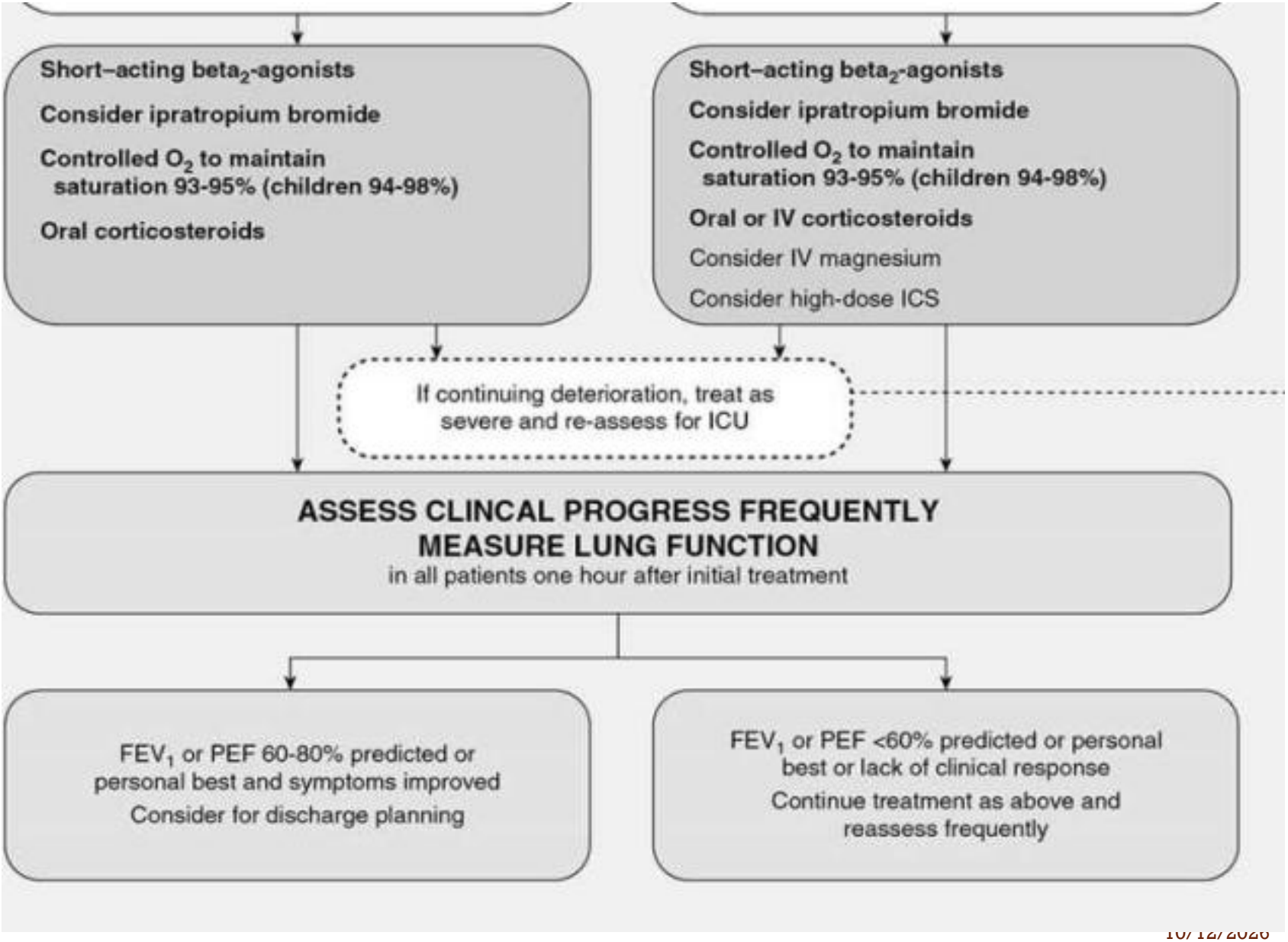
(5) Provide optimal pharmacotherapy with minimal or no adverse effects; and

(6) Meet patients' and families' expectations of and satisfaction with asthma care

- Acute Severe Asthma
  (1) Correct significant hypoxemia;
  (2) Rapidly reverse airflow obstruction; and
  (3) Reduce the likelihood of recurrent severe airflow obstruction

ACUTE SEVERE ASTHMA




Short–acting beta$_2$-agonists
Consider ipratropium bromide
Controlled O_2 to maintain saturation 93-95% (children 94-98%)
Oral corticosteroids
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Consider ipratropium bromide
Controlled O_2 to maintain saturation 93-95% (children 94-98%)
Oral or IV corticosteroids
Consider IV magnesium
Consider high-dose ICS
If continuing deterioration, treat as severe and re-assess for ICU
ASSESS CLINCAL PROGRESS FREQUENTLY
MEASURE LUNG FUNCTION
in all patients one hour after initial treatment
FEV_1 or PEF 60-80% predicted or personal best and symptoms improved
Consider for discharge planning
FEV_1 or PEF <60% predicted or personal best or lack of clinical response
Continue treatment as above and reassess frequently

CHRONIC ASTHMA

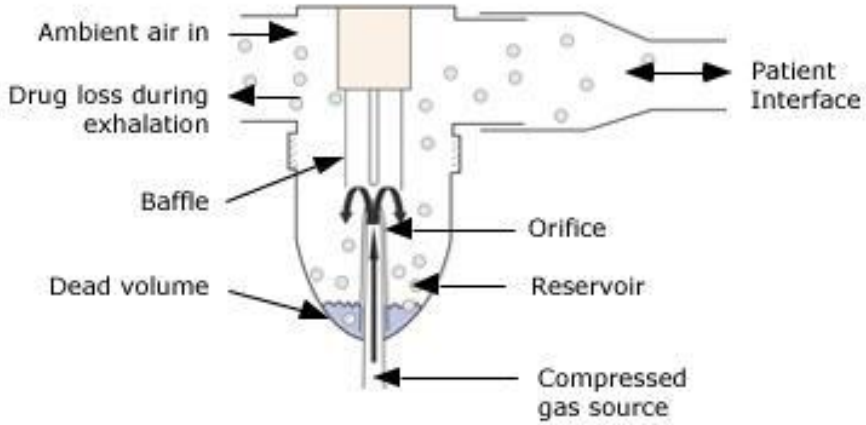
- **Nonpharmacologic Treatment**
  - Environmental control
  - Manage comorbid conditions
  - Self-management skills
    - ✓ Recognize early signs of deterioration
  - Education
    - ✓ Asthma, role of medications, inhalation technique, environmental control, reinforce every visit

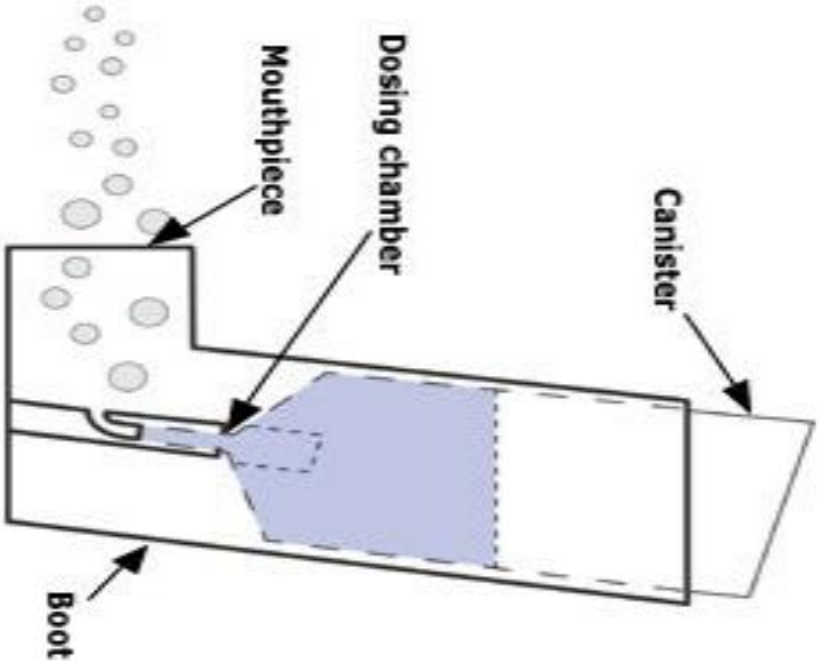
Metered-dose inhaler (MDI)



Nebulizer







PATIENT EDUCATION

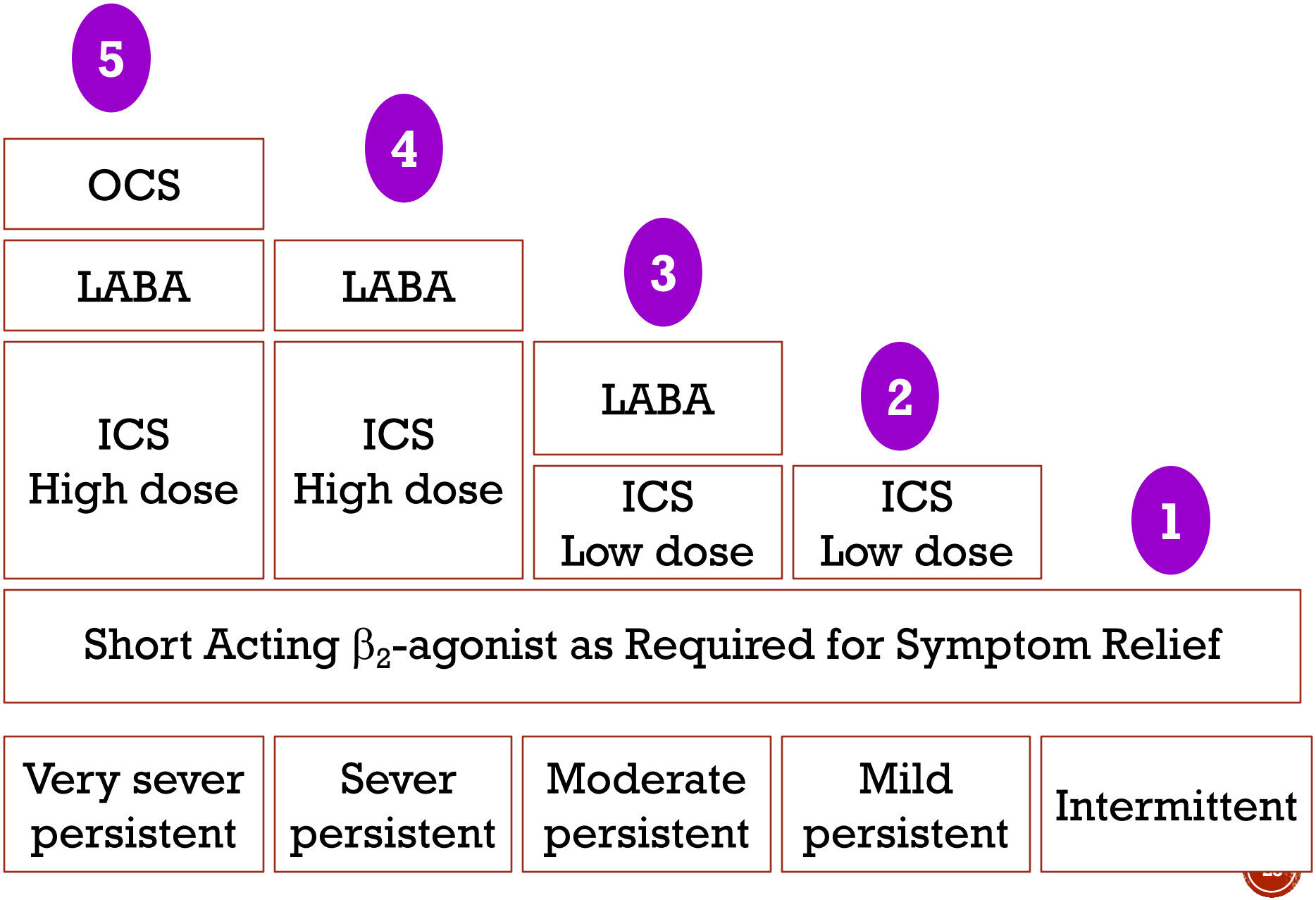
- Appropriate inhalation technique is vital for optimal drug delivery and therapeutic effect
  - up to 30% cannot master MDI technique
- Rinse mouth after inhaled corticosteroids (ICS)
  - < 4 years old usually need to attach a face mask to the inhalation device

General Approach to the Management of Chronic Asthma

Symptom Presentation	Preferred Treatment (Evidence Level)
Symptoms or need for SABA < 2×/mo; no waking due to asthma in last month; and no risk factors for exacerbations, including in prior year	No controller (D)
Infrequent symptoms, but patient has one or more risk factor for exacerbation, eg, low lung function, use of OCS in prior year, intensive care treatment for asthma ever	Low-dose ICS (D)
Symptoms or need for SABA between 2×/mo and 2×/wk, or patient wakes due to asthma ≥ 1 mo	Low-dose ICS (B)
Symptoms or need for SABA > 2×/wk	Low-dose ICS[a] (A)
Troublesome symptoms most days or waking ≥ 1×/wk, especially if any risk factors exist	Medium/high-dose ICS(A) or low-dose ICS/LABA[b] (A)
Symptoms consistent with severely uncontrolled asthma, or with an acute exacerbation	OCS short course AND start of high-dose ICS(A) or moderate-dose ICS/LABA[b] (D)

- ***The presence of one feature of severity is sufficient to place patient in that category***

CLASSIFY SEVERITY Clinical Features Before Treatment	Symptoms	Nocturnal Symptoms	FEV_1 or PEF
STEP 4 Severe Persistent	Continuous Limited physical activity	Frequent	□ 60% predicted Variability > 30%
STEP 3 Moderate Persistent	Daily Attacks affect activity	> 1 time week	60 - 80% predicted Variability > 30%
STEP 2 Mild Persistent	> 1 time a week but < 1 time a day	> 2 times a month	□ 80% predicted Variability 20 - 30%
STEP 1 Mild Intermittent	< 1 time a week Asymptomatic and normal PEF between attacks	< 2 times a month	□ 80% predicted Variability < 20%


5
4
3
2
1
OCS
LABA
LABA
ICS
High dose
ICS
High dose
LABA
ICS
Low dose
ICS
Low dose
Short Acting β_2-agonist as Required for Symptom Relief
Very sever persistent
Sever persistent
Moderate persistent
Mild persistent
Intermittent

INHALED CORTICOSTEROIDS (ICS)

- Use: cornerstone of chronic asthma therapy
  - Improve lung function
  - Reduce severe exacerbations
  - Only therapy shown to reduce risk of asthma death
- Low systemic activity
- Response delayed for several weeks
- Products: beclomethasone dipropionate, budesonide, flunisolide, fluticasone propionate, mometasone furoate, triamcinolone acetonide, ciclesonide

LABAs

- Use: preferred adjunct/ICS combination
  - Adults & most children
  - Better control than increasing ICS dose alone
- Not for quick relief
- Provide long lasting bronchodilation (≥ 12 hours)
- Products
  - Formoterol
  - Salmeterol

- Systemic side effects: dose dependent
- *Not to be used as monotherapy*
  - Increased risk of severe, life threatening exacerbation & asthma related death
  - Preliminary data suggest concomitant ICS may prevent/decrease risk

METHYLXANTHINES

- Mechanism: bronchodilation
  - nonselective *phosphodiesterase* inhibitor
    - Isoenzyme III: airway smooth muscle
    - Isoenzyme IV: inflammatory cell regulation
  - Competitively inhibit adenosine
  - Stimulate catecholamine release
- Use declined due to risk for toxicity
  - Alternative/adjunct therapy

Mast Cell Stabilizer

- Mechanism
  - No bronchodilatory effect
  - Inhibit neurally mediated bronchoconstriction
- Improvement in 1 to 2 weeks
- Alternative to initial ICS therapy but not as effective
- Cromolyn
  - MDI or nebulizer

LEUKOTRIENE MODIFIERS

- Leukotriene receptor antagonists (LTRA)
  - Zafirlukast
  - Montelukast
- 5-lipoxygenase inhibitor
  - Zileuton
- Use: alternative/adjunct therapy
  - Less effective than ICS
  - Oral dosage form
- Adverse effects
  - Hepatic dysfunction

ANTI-IGE: OMALIZUMAB

- Mechanism: recombinant anti-IgE antibody
  - Prevents binding of IgE to mast cells & basophils
    - Decreases release of mediators following allergen exposure
- Use
  - Allergic asthma not well controlled by corticosteroids
  - ≥ ***12 years old***
  - Severe persistent asthma

- Dosage/administration
  - Subcutaneous every 2 to 4 weeks
  - Dosage based on serum IgE level & weight
- Adverse effect
  - Anaphylaxis
    - 70% occur within 2 hours
    - may occur up to 24 hours after injection